Channel Estimation for RIS-Aided mmWave MIMO Systems via One-Sided Sparse Representation

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***Abstract*—In this paper, we investigate the problem of cascaded channel estimation for reconfigurable intelligent surface (RIS)-aided millimeter-wave (mmWave) multiple-input multiple-output (MIMO) systems. To break the accuracy-complexity trade-off in the conventional two-sided sparse representation in the angular domain, herein a one-sided scheme is devised for the cascaded channel. By integrating the dictionary matrices corresponding to the RIS into the coefficient matrix, the cascaded channel is represented using only a single dictionary matrix corresponding to the base station (BS), thereby significantly reducing the complexity. Furthermore, a one-sided block-structure orthogonal matching pursuit (OS-BS-OMP) algorithm is developed, which exploits the inherent block sparsity of the one-sided channel representation. Simulation results demonstrate the superiority of the proposed OS-BS-OMP algorithm over its counterparts.**

***Index Terms*—Channel estimation, reconfigurable intelligent surface, one-sided sparse representation, block sparsity, orthogonal matching pursuit.**

I. Introduction

Along with the widespread application of millimeter-wave (mmWave) technology in the sixth-generation (6G) mobile communications, reconfigurable intelligent surface (RIS) has emerged as a pivotal technology for enhancing signal coverage and spectral efficiency. In RIS-aided wireless communication systems, acquiring the cascaded channel state information (CSI) from the base station (BS) to the RIS, and then to the user equipment (UE), is a fundamental prerequisite for achieving signal enhancement and high beamforming gain [1], [2]. However, due to the high dimensionality of the cascaded channel and lack of baseband-signal processing capabilities at the RIS units, achieving high-precision channel estimation for the RIS-aided systems within limited pilot overhead remains a significant challenge and a focus of current research.

Conventionally, the least squares (LS) [3], [4] or the linear minimum mean-squared-error (LMMSE) [5], [6] estimator is used for channel estimation. However, these methods overlook the angular sparsity of mmWave channels and require prohibitive pilot overhead. To address this issue, subsequent studies have tried to estimate the cascaded channel of the RIS-aided communication systems from its parametric sparse model. It is known that the maximum likelihood estimation (MLE) is optimal [7] and can be performed via the manifold optimization (MO) technique [8], while it depends on the often-unavailable channel rank information. As a substitute, the low-rank matrix recovery [9], which aims to minimize the nuclear norms of these channel matrices, has been employed. In addition, the parallel factor (PARAFAC) tensor model has been utilized in [10] to achieve the similar objective. However, these methods require mutually orthogonal time-domain pilots, inevitably degrading their performance.

Recently, the compressed sensing (CS) techniques, such as the cascaded orthogonal matching pursuit (Cascaded OMP) [11] and the alternating block-structured (BS)-OMP [12] algorithms, have been widely applied to the RIS-aided channel estimation. Among them, the BS-RIS and RIS-UE subchannels are respectively represented in the angular domain in a sparse way. However, there exists the trade-off between the estimation accuracy and the dictionary resolution. Even worse, the massive size of the planar RIS array incurs the prohibitively large dimensions of the dictionary matrices. To mitigate the basis mismatch in such a two-sided sparse representation, an atomic norm minimization scheme is proposed in [13], which achieves the super-resolution estimation in the continuous domain. Nevertheless, this approach typically gets involved with the high-dimensional semi-definite programming (SDP), imposing a substantial computational burden.

To alleviate the high complexity of the two-sided representation in the angular domain, a new one-sided representation scheme is proposed in this paper for the cascaded channel of the RIS-aided systems. Unlike the existing two-sided representation, the dictionary matrices corresponding to the RIS are integrated into the coefficient matrix in the one-sided angular-domain representation. As a result, the cascaded channel is represented with only a single dictionary matrix corresponding to the BS, which saves the complexity by a large degree. Furthermore, by exploiting the block sparsity, we propose a one-sided block-structure orthogonal matching pursuit (OS-BS-OMP) algorithm for the estimation of the cascaded channel. Simulation results demonstrate that the proposed algorithm achieves the superior performance of estimation accuracy.

II. Signal Model

Consider an RIS-aided mmWave system, which consists of a BS equipped with N transmit antennas, a RIS with M reflecting elements, and a single-antenna user. Here, the BS transmit antennas and the RIS reflecting elements are arranged

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as a uniform linear array (ULA) and a uniform planar array (UPA), respectively. In this paper, the inter-antenna spacing is set as a typical value, that is half a wavelength. Suppose that before data transmission, the channel remains quasi-static within a single training period consisting of T consecutive time slots. During the phase of channel training, the BS transmits the pilot vectors, $\mathbf{x}_1$, $\mathbf{x}_2$, $\cdots$, $\mathbf{x}_T$, to the receiver time slot by time slot. At the end of this process, the CSI is estimated at the receiver based on the collected observations. Then, the estimated CSI is fed back to the transmitter so that the data to be transmitted can be precoded in a spectrally efficient way.

The reflecting coefficients of the RIS vary from time slot to time slot. In the practical deployment, considering the hardware cost and control complexity, the phase shift of each RIS reflecting element is typically restricted to a finite set of discrete values. Let b denote the number of quantization bits. The total number of available discrete phase levels is given by $K = 2^b$. Accordingly, the set of the candidate values for each reflecting coefficient is represented as [14]

$$\mathcal{F} = \{1, e^{\jmath\Delta\theta}, \ldots, e^{\jmath(K-1)\Delta\theta}\}, \tag{1}$$

where $\Delta\theta = 2\pi/K$ is the step size of the phase quantization. Consequently, at the tth time slot, each entry of the reflecting coefficient vector $\mathbf{v}_t = [v_{t,1}, v_{t,2}, \ldots, v_{t,M}]^T$ satisfies the constraint of $v_{t,m} \in \mathcal{F}$, $m = 1, 2, \ldots, M$.

Based on the aforementioned channel estimation protocol, the user's received signal at the tth time slot, denoted by y_t, is expressed as:

$$y_t = \mathbf{x}_t^H \mathbf{G} \mathbf{V}_t \mathbf{h} + n_t, \tag{2}$$

where $\mathbf{G} \in \mathbb{C}^{N\times M}$ and $\mathbf{h} \in \mathbb{C}^M$ denote the fading channels from the BS to the RIS and from the RIS to the UE, respectively; $\mathbf{V}_t = \mathcal{D}(\mathbf{v}_t)$ denotes the phase-shift matrix of the RIS at the tth time slot; and n_t is the additive Gaussian white noise following an independent and identically distributed (i.i.d.) process. Further, y_t can be expressed as

$$y_t = \mathbf{x}_t^H \mathbf{G} \cdot \mathcal{D}(\mathbf{h}) \cdot \mathbf{v}_t + n_t. \tag{3}$$

Define the state information of the cascaded channel of the massive MIMO system as

$$\mathbf{H}_c = \mathbf{G} \cdot \mathcal{D}(\mathbf{h}). \tag{4}$$

Once accurate information about $\mathbf{H}_c$ is obtained, one can design BS precoding and RIS reflection coefficients to achieve high spectral efficiency. Therefore, in this paper, it is aimed to estimate the transformation coefficient matrix $\mathbf{H}_c$ in a statistically efficient manner.

III. Algorithm Development

A. One-Sided Representation of $\mathbf{H}_c$

In the mmWave band, it is known that the channel between the BS and RIS, denoted by $\mathbf{G}$, can be well described by the Saleh-Valenzuela model [15]–[17]:

$$\mathbf{G} = \sum_{l=1}^{L} \rho_l \mathbf{a}_t(\phi_l) \mathbf{a}_r^H(\theta_l, \gamma_l), \tag{5}$$

where L denotes the number of paths, ρ_l represents the complex gain of the lth path, following a Gaussian distribution, whereas ϕ_l and θ_l (γ_l) are the angle of departure (AoD) and the azimuth (elevation) angle of arrival (AoA), respectively. The vectors $\mathbf{a}_t(\phi_l)$ and $\mathbf{a}_r(\theta_l, \gamma_l)$ represent the normalized transmit and receive array responses at the AoD ϕ_l and the AoA (θ_l, γ_l), respectively. Given that the BS is equipped with N transmit antennas and the RIS consists of M ($M = M_x M_y$) reflecting elements, then, $\mathbf{a}_t(\phi_l)$ can be expressed as:

$$\mathbf{a}_t(\phi_l) = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & e^{\jmath\pi\cos(\phi_l)} & \cdots & e^{\jmath\pi(N-1)\cos(\phi_l)} \end{bmatrix}^T, \tag{6}$$

and

$$\mathbf{a}_r(\theta_l, \gamma_l) = \mathbf{a}_x(\mu_l) \otimes \mathbf{a}_y(\nu_l), \tag{7}$$

respectively, with

$$\mathbf{a}_x(\mu_l) \triangleq \frac{1}{\sqrt{M_x}} \begin{bmatrix} 1 & e^{\jmath\mu_l} & \cdots & e^{\jmath(M_x-1)\mu_l} \end{bmatrix}^T, \tag{8}$$

$$\mathbf{a}_y(\nu_l) \triangleq \frac{1}{\sqrt{M_y}} \begin{bmatrix} 1 & e^{\jmath\nu_l} & \cdots & e^{\jmath(M_y-1)\nu_l} \end{bmatrix}^T, \tag{9}$$

and $\mu_l \triangleq \pi\cos(\gamma_l)$, $\nu_l \triangleq \pi\sin(\gamma_l)\cos(\theta_l)$. Similarly, $\mathbf{h}$ is modeled in the millimeter-wave band as:

$$\mathbf{h} = \sum_{l=1}^{L'} \alpha_l' \mathbf{a}_r(\theta_l', \gamma_l') = \mathbf{A}_R' \boldsymbol{\alpha}, \tag{10}$$

where L' denotes the number of the paths from the RIS to the UE, and θ_l' (γ_l') is the azimuth (elevation) AoD of the lth path. In addition, $\boldsymbol{\alpha} \triangleq \left[\alpha_1', \alpha_2', \cdots, \alpha_{L'}'\right]^T$ represents the vector of the path gains, and it is defined that,

$$\mathbf{A}_R' = [\mathbf{a}_r(\theta_1', \gamma_1')\ \mathbf{a}_r(\theta_2', \gamma_2') \cdots \mathbf{a}_r(\theta_{L'}', \gamma_{L'}')] \in \mathbb{C}^{M\times L'}. \tag{11}$$

Within the sparse recovery framework, the signal model of (2) is usually reconstructed based on the two-sided sparse representation of $\mathbf{H}_c$ [11]. Due to the high dimensionality of such a representation, the number of unknown coefficients to be estimated is enormous, that is equal to the product of the numbers of the coefficients in the sparse representations of $\mathbf{G}$ and $\mathbf{h}$. To reduce the computational complexity, it is proposed in this paper to reformulate the signal model of (2) with the one-sided representation of $\mathbf{H}_c$ in the angular domain as a substitute scheme.

Normally, the nature of the mmWave band imposes a small value onto L of (5). This provides the possibility to represent $\mathbf{G}$ in a sparse way. More specifically, it is feasible to define a discrete set of $\boldsymbol{\varphi} = \{\varphi_1, \cdots, \varphi_{N_G}\}$ ($N_G \gg L$) at the evenly spaced points in $[0^\circ,\ 180^\circ]$, which contains N_G virtual AoDs. Furthermore, it is assumed that $\boldsymbol{\varphi}$ is sufficiently dense such that the true AoDs are all on-grid. This means that $\mathbf{G}$ can be sparsely represented in the angular domain. Conventionally, the cascaded channel is represented in a two-sided way within the framework of sparse signal recovery, such as in [11] and [12], which uses the two dictionary matrices corresponding to the BS and RIS. As a different scheme, the one-sided representation approach is developed in this paper, where $\mathbf{H}_c$ is sparsely expressed only at the side of the BS. That is,

$$\mathbf{G} = \sum_{n=1}^{N_G}\sum_{l=1}^{L} \sigma_{n,l}\mathbf{a}_t(\varphi_n)\mathbf{a}_r^H(\theta_l,\gamma_l) \triangleq \mathbf{A}_L\boldsymbol{\Sigma}\mathbf{A}_R^H, \tag{12}$$

where $\mathbf{A}_L$ and $\mathbf{A}_R$ stand for the dictionary matrices at the transmitting and receiving sides, respectively, as

$$\mathbf{A}_L = \begin{bmatrix}\mathbf{a}_t(\varphi_1) & \mathbf{a}_t(\varphi_2) & \cdots & \mathbf{a}_t(\varphi_{N_G})\end{bmatrix} \in \mathbb{C}^{N\times N_G}, \tag{13}$$

$$\mathbf{A}_R = [\mathbf{a}_r(\theta_1,\gamma_1)\ \mathbf{a}_r(\theta_2,\gamma_2)\cdots\ \mathbf{a}_r(\theta_L,\gamma_L)] \in \mathbb{C}^{M\times L} \tag{14}$$

and $\boldsymbol{\Sigma} = [\sigma_{n,l}] \in \mathbb{C}^{N_G\times L}$ consists of the coefficients associated with the $N_G L$ components of (12). It is noted that $\mathbf{A}_L$ is an overcomplete dictionary matrix defined in the angular set of $\boldsymbol{\varphi}$. Compare (5) and (12), and we have

$$\sigma_{n,l} = \begin{cases}\rho_l, & \text{if } \varphi_n = \phi_l;\\ 0, & \text{otherwise.}\end{cases} \tag{15}$$

This implies that $\boldsymbol{\Sigma}$ is column-sparse, i.e., each column of $\boldsymbol{\Sigma}$ is a sparse vector with a small number of nonzero entries.

Substituting (12) for $\mathbf{G}$ in (2), it is derived that,

$$\begin{aligned} y_t &= \mathbf{x}_t^H\mathbf{A}_L\boldsymbol{\Sigma}\mathbf{A}_R^H\mathbf{V}_t\mathbf{h} + n_t\\ &= \mathbf{x}_t^H\mathbf{A}_L\boldsymbol{\Sigma}\big(\mathbf{A}_R^H \odot \mathbf{h}^T\big)\mathbf{v}_t + n_t\\ &= \mathbf{x}_t^H\mathbf{A}_L\mathbf{W}\mathbf{v}_t + n_t, \end{aligned} \tag{16}$$

where

$$\mathbf{W} \triangleq \boldsymbol{\Sigma}\big(\mathbf{A}_R^H \odot \mathbf{h}^T\big) \in \mathbb{C}^{N_G\times M} \tag{17}$$

is termed as the transformed coefficient matrix. Since there exist only L paths in the BS-RIS channel, the matrix $\boldsymbol{\Sigma}$ contains at most L nonzero rows. Consequently, at least $N_G - L$ rows of $\mathbf{W}$ are zero. In the angular domain, the cascaded channel can therefore be expressed as

$$\mathbf{H}_c = \mathbf{A}_L\mathbf{W}. \tag{18}$$

As a result, the estimation of the cascaded channel $\mathbf{H}_c$, is converted into that of the transformed coefficient matrix $\mathbf{W}$. Since $\mathbf{H}_c$ is expressed here with only the dictionary matrix corresponding to the BS, that is $\mathbf{A}_L$, Eq. (18) is termed as the one-sided representation of $\mathbf{H}_c$.

B. Channel estimation with the OS-BS-OMP

By imposing the vectorization operation onto the right-hand side of (16), it is derived that,

$$y_t = \big(\mathbf{v}_t^T \otimes (\mathbf{x}_t^H\mathbf{A}_L)\big)\mathbf{s} + n_t \triangleq \boldsymbol{f}_t^T\mathbf{s} + n_t, \tag{19}$$

where $\boldsymbol{f}_t^T$ is defined as $\boldsymbol{f}_t^T = \mathbf{v}_t^T \otimes (\mathbf{x}_t^H\mathbf{A}_L)$, and $\mathbf{s} = \mathrm{vec}(\mathbf{W})$ is constructed by stacking the columns of $\mathbf{W}$ into a vector. Considering the one-sided representation of $\mathbf{H}_c$ derived in (18), each nonzero row of $\mathbf{W}$ corresponds to an existing transmitting direction from the BS. Therefore, the vectorized coefficient vector $\mathbf{s}$ exhibits the block-structure sparsity. In detail, we divide $\mathbf{s}$ into M blocks, and the mth block is the mth column of $\mathbf{W}$, that is, $\mathbf{s}_m = \mathbf{W}(:,m)$, for $m = 1,2,\cdots,M$. The block sparsity of $\mathbf{s}$ indicates that for each block indexed by m, i.e., $\mathbf{s}_m$, only a limited number of elements are nonzero. Such block sparsity is directly related with the number of the propagation paths of $\mathbf{G}$, i.e., L. This means that there are at least $N_G - L$ entries of $\mathbf{s}_m$ being zero.

Algorithm 1 One-Sided Block-Structured OMP (OS-BS-OMP)

Require: $\mathbf{y}_T$, $\mathbf{F}_T \in \mathbb{C}^{T\times(N_G M)}$, M, N_G.
Ensure: Estimated coefficient vector, $\mathbf{s}_{est}$.
1: **Initialization:** $\mathbf{r}_0 = \mathbf{y}_T, \Omega = \emptyset, i = 1, \boldsymbol{\Phi} \triangleq \mathbf{F}_T$;
2: **Normalization:** $\boldsymbol{\phi}_j = \boldsymbol{\phi}_j/\|\boldsymbol{\phi}_j\|_2, \forall j \in \{1,\ldots,N_G M\}$;
3: **while** $i \le M$ **do**
4: **Selection of relevant blocks:**
$b_{opt} = \arg\max_{b\in\{1,\ldots,M\}} \|\boldsymbol{\Phi}[b]^{\mathrm{H}}\mathbf{r}_{i-1}\|_2$;
5: **Neighboring block suppression:**
$\boldsymbol{\Phi}[b] = \mathbf{O}_{T\times N_G}, \forall b \in \{b_{opt}-1, b_{opt}, b_{opt}+1\}$;
6: **Identification of dominant basis vectors:**
$[\mathbf{x}_{local}, \Gamma_{local}] = \mathrm{OMP}(\mathbf{F}_T[b_{opt}], \mathbf{r}_{i-1})$;
7: **Support update:**
$\Omega = \Omega \cup \{\Gamma_{local} + (b_{opt}-1)M\}$;
8: **Residual update:**
$\mathbf{r}_i = \mathbf{r}_{i-1} - \mathbf{F}_T[b_{opt}]\mathbf{x}_{local}$;
9: **Convergence check:**
if $\big|\|\mathbf{r}_{i-1}\|_2 - \|\mathbf{r}_i\|_2\big| < 0.1\|\mathbf{y}_T\|_2$ **break**;
10: **Index update:** $i = i+1$;
11: **end while**
12: **Refinement:**
13: $\mathbf{s}_{est}(\Omega) = \big(\mathbf{F}_T^{\mathrm{H}}(:,\Omega)\mathbf{F}_T(:,\Omega)\big)^{-1}\mathbf{F}_T^{\mathrm{H}}(:,\Omega)\mathbf{y}_T$.
14: **return** $\mathbf{s}_{est}$;

Concatenate $\{y_t\}_{t=1}^T$ together, and the collected observations at the Tth time slot is expressed as:

$$\mathbf{y}_T = \mathbf{F}_T\mathbf{s} + \mathbf{n}_T, \tag{20}$$

where $\mathbf{F}_T = [\mathbf{f}_1\ \mathbf{f}_2\ \cdots\ \mathbf{f}_T]^T$, $\mathbf{y}_T = [y_1\ y_2\ \cdots\ y_T]^T$, and $\mathbf{n}_T = [n_1\ n_2\ \cdots\ n_T]^T$. Now the work of this paper becomes the estimation of the coefficient vector of $\mathbf{s}$, and further $\mathbf{W}$, from the signal model of (20). Once this is done, the full information of the cascaded channel $\mathbf{H}_c$ can be accordingly recovered from (18).

As a tool to utilize the block sparsity of $\mathbf{s}$, and to recover the coefficient vector from $\mathbf{y}_T$ in a statistically efficient way, the OS-BS-OMP algorithm is devised in this paper. Overall, the blocks which are the most relevant to the observation of $\mathbf{y}_T$ are firstly selected one by one. Once a certain block is selected, a neighboring block suppression (NBS) mechanism is introduced to decrease the trial numbers of block selection, and to avoid redundant computation. Then, the orthogonal matching pursuit (OMP) is adopted to identify and reserve the dominant basis vectors of each selected block, producing the corresponding coefficient estimates, $\hat{\mathbf{s}}_i$. Finally, the OMP is re-run to finalize the coefficient estimates for the dominant basis vectors of all the selected blocks. The implementation steps are summarized in Algorithm 1.

TABLE I: CPU Time of the Cascaded Channel Estimation (s)

Pilot (Path) / Algorithm	$T = 100$ ($L = 10$)	$T = 100$ ($L = 40$)	$T = 200$ ($L = 10$)	$T = 200$ ($L = 40$)
JSENNM-CE	67.10	65.91	71.22	73.17
A-OMP	30.37	33.61	14.62	17.04
C-OMP	0.54	0.58	0.73	0.77
OS-BS-OMP	0.04	0.04	0.06	0.08

IV. Simulation Results

In this section, a set of numerical simulations are carried out to evaluate the performance of the proposed OS-BS-OMP algorithm. In detail, the cascaded channel estimation results of [11], [12], and [9] are offered for comparison, which are labeled as "C-OMP", "A-OMP", and "JSENNM-CE", respectively. We consider the RIS-aided mmWave system with $N = 16$ antennas at the BS and the RIS consisting of $M = 8 \times 8 = 64$ elements. The grid size of $\mathbf{A}_L$ is set as $N_G = 181$; and for the methods of C-OMP, A-OMP, and JSENNM-CE, $\mathbf{A}_R$ and $\mathbf{A}'_R$ are represented in the 2-D grid sized with $(M_{G,x}, M_{G,y}) = (32, 32)$. The number of scattering paths for the BS-RIS channel is set as $L = 10$ or $L = 40$, while the RIS-UE channel has $L' = 10$ paths. The path gains follow a complex Gaussian distribution $\mathcal{CN}(0, 1)$, and the performance is measured in terms of the normalized mean squared error (NMSE) of $\mathbf{H}_c$, which is defined as

$$\mathrm{NMSE} = \mathbb{E}\left[\frac{\|\mathbf{H}_{c,est} - \mathbf{H}_c\|_F^2}{\| \mathbf{H}_c \|_F^2}\right]. \tag{21}$$

Fig. 1 illustrates the NMSE performance versus SNR for the pilot length of $T = 100$. It is observed that the proposed OS-BS-OMP algorithm provides the much lower NMSE than the C-OMP and A-OMP methods, particularly in the low SNR regime ranging from -20 dB to 0 dB. This is because the one-sided representation is utilized in our method, which avoids the off-grid error of $\mathbf{A}_R$ and $\mathbf{A}'_R$ at the side of the RIS. It is also seen that the JSENNM-CE is inferior to the proposed method due to the loss of the pilot orthogonality. Even as the number of paths increases from $L = 10$ to $L = 40$, the proposed algorithm maintains a stable performance advantage, demonstrating its robustness against increased channel complexity.

In Fig. 2, we examine the NMSE performance with an increased pilot length of $T = 200$. As expected, the estimation accuracy of all the algorithms improves with the increased training overhead. Moreover, the proposed OS-BS-OMP still provides the highest precision across the entire SNR range from -20 dB to 5 dB. This means that compared with the conventional methods using the two-sided representation style, the proposed algorithm can achieve the desired estimation accuracy with significantly less training overhead.

Table I shows the CPU time of the different methods for the cascaded channel estimation. Unlike the methods of the C-OMP and the A-OMP, which get involved with the two-sided representation of $\mathbf{H}_c$, and suffer the computational burden from the huge-scale signal recovery, our one-sided representation provides only one light dictionary matrix to $\mathbf{H}_c$. As a result, the computational complexity of our method is obviously smaller than the other ones, which is consistent with the motivation of our work.

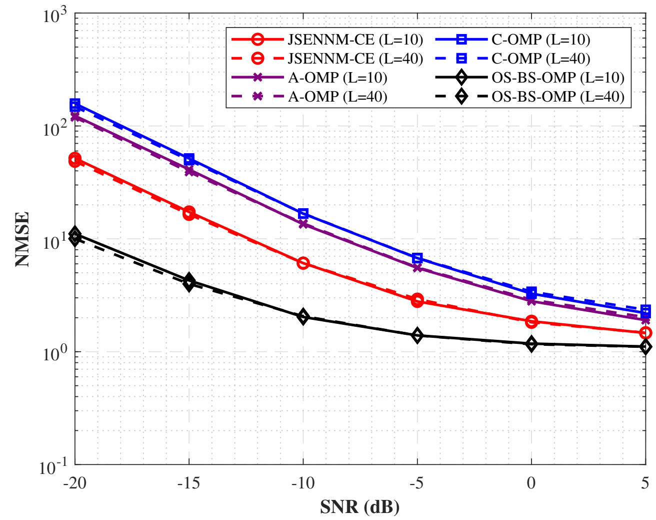

Fig. 1: NMSEs of the estimation of the cascaded channel, $T = 100$.

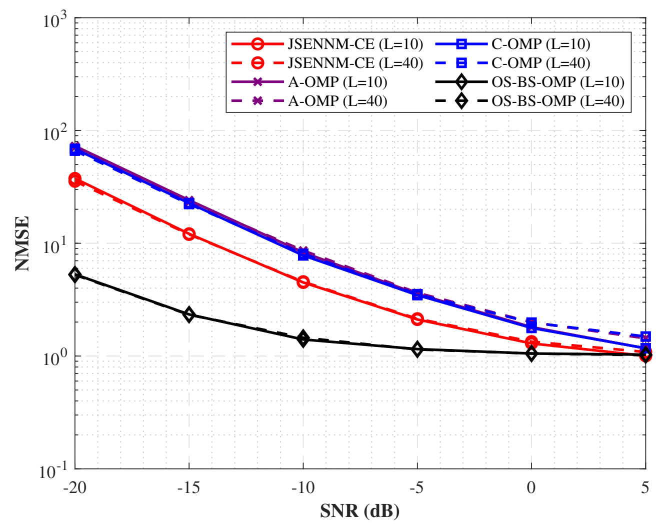

Fig. 2: NMSEs of the estimation of the cascaded channel, $T = 200$.

V. Conclusion

In this paper, a new one-sided sparse representation approach is established in the angular domain for the acquisition of the CSI in the RIS-aided mmWave communication, to alleviate the high complexity of the conventional two-sided representation method. Accordingly, the OS-BS-OMP algorithm is devised to achieve the estimation of the cascaded channel. Simulation results show that the proposed method can provide higher estimation accuracy with lower computational complexity. In the future, the multi-user scenario will be further explored.

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